SPECIAL SESSION PROPOSAL

6th International Conference on Emerging Electronics and Automation (E2A 2027)

National Institute of Technology Silchar | Jan 20-22, 2027 | In Person

AI/ML- Driven Control and Navigation for Autonomous UAV Systems

Acronym: SS-DRONES 2027

Special Session Title	AI/ML- Driven Control and Navigation for Autonomous UAV Systems
Acronym	SS-DRONES 2027
Proposer	Dr. Vasanthakumar Sekar, Assistant Professor (Sr.), VIT University, Chennai, India. vasanthakumar.s@vit.ac.in
Corresponding Proposer	Prof. K. Srinivasan, Professor and Head, Dept. of ICE, NIT Tiruchirappalli, India. srinikkn@nitt.edu

1. Special Session Title and Acronym

Title	AI/ML- Driven Control and Navigation for Autonomous UAV Systems
Acronym	SS-DRONES 2027

2. Brief Description of the Special Session

Autonomous Unmanned Aerial Vehicles are rapidly transforming several domains, including surveillance, infrastructure inspection, disaster management, precision agriculture, logistics, defence, smart cities, and environmental monitoring. Recent advances in artificial intelligence, machine learning, embedded computing, sensing, and automation have enabled UAVs to perform complex missions with improved autonomy, adaptability, and reliability. However, safe and intelligent UAV operation still faces major challenges, particularly in uncertain, dynamic, cluttered, and GPS-denied environments.

This special session focuses on recent developments in AI/ML-driven control, navigation, perception, and decision-making for autonomous UAV systems. The session aims to bring together researchers, academicians, industry professionals, and practitioners working on intelligent aerial systems, robust flight control, autonomous navigation, sensor fusion, computer vision, reinforcement learning, fault-tolerant control, and real-world UAV deployment.

A key emphasis of this session is on autonomous UAV operation under practical constraints such as wind disturbances, actuator uncertainty, communication limitations, obstacle-rich environments, limited onboard computation, and loss or degradation of GPS signals. Topics may include AI-assisted flight control, machine-learning-based trajectory planning, GPS-denied navigation, vision-based localization, SLAM, swarm coordination, adaptive and robust control, fault detection and recovery, edge intelligence, and UAV applications in real-world scenarios.

The session will provide a focused platform for discussing innovative algorithms, experimental validations, simulation frameworks, hardware implementation, and field-level deployment of UAV technologies. It will encourage interdisciplinary research combining control engineering, robotics, electronics, communication, artificial intelligence, machine learning, and cyber-physical systems.

By organizing this special session, the conference will promote high-quality research contributions addressing the next generation of autonomous UAV systems. The session is expected to support knowledge exchange, foster collaboration, and highlight emerging trends in intelligent drone technologies for safe, reliable, and impactful real-world applications.

3. Name, Affiliation, Email Address and Short CV of the Proposer

Dr. Vasanthakumar Sekar

Name: Dr. Vasanthakumar Sekar
Affiliation: Assistant Professor (Sr.), School of Computer Science and Engineering, VIT University, Chennai, India
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LinkedIn: www.linkedin.com/in/vasanthakumar-sekar-48b92565
Research Area: UAV control design, flight dynamics, robust/adaptive control, autonomous aerial systems, SIL/HIL validation

Short CV: Dr. Vasanthakumar Sekar is an Assistant Professor (Sr.) in Computer Science and Engineering at VIT University, Chennai, India, and a control system engineer with strong research and industrial experience in UAV control design, flight dynamics, advanced control systems and simulation. He obtained his Ph.D. in Control Design Engineering of UAV Systems from the National Institute of Technology Tiruchirappalli. His academic background also includes an M.S. in Control Engineering from Coventry University, UK, and an M.Tech. in Control and Instrumentation Engineering from Thiagarajar College of Engineering, Madurai.

His technical expertise includes MATLAB/Simulink, Python, LabVIEW, UAV control, adaptive and robust control, optimization, autopilot systems, embedded controllers, Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) testing. He worked as a Flight Dynamics and Control Engineer at Sarla Aviation, Bangalore, where he developed and validated 6-DOF nonlinear dynamic models for hybrid UAV platforms, designed and tuned INDI, ADRC and PID-based cascaded flight control loops, performed Monte Carlo simulations under disturbances and uncertainty, and integrated control algorithms with Veronte Autopilot and X-Plane for simulation and validation.

His research publications include works on quadcopter control, event-triggered optimal robust control, centralized cascade control, trajectory tracking and UAV guidance. He also holds a DGCA

Drone Pilot License for small and medium category drones, which strengthens the practical relevance of his proposed special session.

4. List of Potential Reviewers:

The following list provides suggested potential Reviewers names, affiliations and emails.

S. No.	Potential Reviewers	Affiliation	Email
1	**Prof. K. Srinivasan**	Professor and Head, Dept. of ICE, NIT Tiruchirappalli	srinikkn@nitt.edu
2	**Dr. Avinash Kumar Roy**	System Engineer, AIRBUS, Bangalore	avnkroy0707@gmail.com
3	**Dr Sridevi Veerasingam**	Assistant Professor at National Institute of Technology, Tiruchirappalli	sridevi@nitt.edu
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6	**Mr. Ajil Roy**	Team Manager, Alstom, Bangalore Urban, Karnataka, India	ajil.roy@outlook.com
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10	**Dr. Jesu Godwin Devaraj**	Postdoctoral Researcher, İstinye Üniversitesi, Istanbul, Türkiye	djesugodwin@gmail.com
11	**Mr. Anand Kumar**	Researcher at National Institute of Technology, Tiruchirappalli	manandonline@gmail.com
12	**Dr. Sivanathan Kandhasamy**	Associate Professor, SRM Institute of Science and Technology (SRMIST)	sivanaathan.k@gmail.com
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14	**Dr. Janeshwaran G**	Assistant Professor, School of Electrical Engineering (Instrumentation), VIT Vellore	janeshg96@gmail.com
15	**Dr. Raghavendran S**	Assistant Professor, Shanmugha Arts, Science, Technology & Research Academy (SASTRA), Thanjavur	raghavagi@gmail.com
16	**Dr. Krishnasamy Subramaniyan**	Associate Professor, Dayananda Sagar College of Engineering, Bangalore	krushnasamy@yahoo.co.in
17	**Dr. Amit Pal**	Nexteer Automotive, Bengaluru, Karnataka, India	amit91pal@outlook.com
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19	**Dr. Balaji C**	Assistant Professor, SRM University, Chennai	balajic2003@gmail.com
20	**Dr. P. Suresh Kumar**	Assistant Professor, VIT Vellore	suresh.204850@gmail.com

5. Rationale for Organizing this Special Session

Drones and autonomous UAV systems have become intelligent cyber-physical platforms that integrate sensing, control, communication, computation, AI/ML, and automation. This makes the proposed session highly relevant to E2A 2027, which focuses on emerging electronics and automation.

Drone-related research is often spread across different tracks such as control, robotics, AI, communication, embedded systems, and applications. A dedicated special session will bring these works together and provide a focused platform for technical discussion and collaboration.

The session is also important due to the increasing use of UAVs in agriculture, infrastructure inspection, disaster response, surveillance, delivery, environmental monitoring, and smart-city applications. These applications require reliable control, intelligent navigation, robust sensing, secure communication, and validation through simulation, SIL/HIL, and field trials.

The proposed session will encourage contributions from academia and industry on practical UAV challenges such as GPS-denied navigation, wind disturbance, actuator uncertainty, onboard computation limits, safety, reliability, and cybersecurity. The proposer's background in UAV control design, nonlinear modelling, robust control, simulation, SIL/HIL validation, and drone operations will help coordinate a technically strong and impactful session.

6. Proposed Technical Scope and Drone Application Themes

Technical Topics	Application Domains
• UAV modelling, flight dynamics, stability analysis and aerodynamic characterization • Guidance, navigation and control for multirotor, fixed-wing and hybrid VTOL UAVs • AI/ML-driven flight control, adaptive control and learning-based autonomy • Robust, fault-tolerant and resilient control under wind disturbance and actuator uncertainty • GPS-denied navigation, vision-based localization and inertial navigation systems • Autonomous mission planning, trajectory tracking, path planning and obstacle avoidance • Computer vision, deep learning, SLAM and sensor fusion for drone perception • Reinforcement learning and intelligent	• Precision agriculture and crop health monitoring • Aerial mapping, remote sensing and geospatial analytics • Power-line, pipeline, bridge and infrastructure inspection • Disaster response, search and rescue, and emergency logistics • Healthcare delivery, medical supply transport and remote access support • Environmental monitoring, forest/fire surveillance and smart-city services • Traffic monitoring, crowd surveillance and public safety applications • Defense, border monitoring and tactical surveillance missions • Industrial automation, warehouse monitoring and factory inspection • Marine, coastal and flood monitoring using

decision-making for autonomous UAVs • Swarm drones, cooperative control, formation flight and multi-agent coordination • Human–drone interaction, remote supervision and autonomous mission management • Drone communications, IoT, 5G/6G, edge intelligence and cloud-connected aerial systems • Cybersecurity, secure communication and safety assurance for UAV networks • Digital twin, SIL/HIL simulation and real-time validation of UAV control systems • Energy-aware control, battery management and efficient propulsion systems • Payload integration, onboard embedded systems and real-time avionics implementation	aerial robotic systems • Wildlife monitoring, biodiversity assessment and conservation support • Urban air mobility, eVTOL systems and future aerial transportation • Mining, construction-site monitoring and hazardous-area inspection • Delivery drones, last-mile logistics and autonomous cargo transport • Educational, research and training platforms for UAV system development

7. Alignment with E2A 2027

The proposed special session “AI/ML-Driven Control and Navigation for Autonomous UAV Systems” is strongly aligned with the scope of E2A 2027, as it connects emerging electronics, automation, control systems, artificial intelligence, sensing, communication, and real-world engineering applications.

Autonomous UAVs require advanced electronic systems, embedded controllers, sensors, communication modules, AI/ML algorithms, and automated decision-making methods. Therefore, this topic directly supports the conference theme of emerging technologies in electronics and automation.

The session will encourage research contributions on UAV control, GPS-denied navigation, intelligent sensing, perception, embedded implementation, simulation, SIL/HIL validation, and practical drone applications. These areas are highly relevant to the interdisciplinary objectives of E2A 2027 and will attract researchers from control engineering, robotics, AI, electronics, instrumentation, and automation domains.

By focusing on intelligent and deployable UAV technologies, this special session will add practical and research value to E2A 2027 and support discussions on next-generation autonomous aerial systems.

8. Expected Session Outcomes

The proposed special session is expected to bring together researchers, academicians, industry experts, and practitioners working on intelligent UAV technologies. It will provide a focused platform to present recent advances in AI/ML-driven control, autonomous navigation, sensing, and real-world drone applications.

The session is expected to generate the following outcomes:

- High-quality research contributions on autonomous UAV control, GPS-denied navigation, perception, sensing, and intelligent decision-making.
- Focused technical discussions on current challenges such as wind disturbances, actuator uncertainty, onboard computation limits, safety, reliability, and field deployment.
- Knowledge exchange between academia and industry for developing practical and deployable UAV solutions.
- Identification of emerging research directions in AI/ML-based drone autonomy, sensor fusion, SLAM, robust control, and swarm UAV systems.
- Promotion of real-world UAV applications in agriculture, infrastructure inspection, disaster response, surveillance, mapping, smart cities, and environmental monitoring.

Overall, the session will support E2A 2027 by strengthening interdisciplinary research discussions on next-generation autonomous aerial systems and their practical impact.